Supplementary Material

Validator-Backed Auditable Robust Aggregation for Secure Networked Federated Learning

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Supplementary Table S1. Trusted-root and FLAME-style extension baselines

Table S1 reports extension baselines that use assumptions different from the main no-root-data robust aggregation setting. FLTrust uses a trusted server-side root set with 100 samples to construct a reference update. The FLAME-style baseline uses cosine-distance clustering, clipping, and calibrated Gaussian noise; it is reported as a FLAME-style implementation rather than a byte-for-byte reproduction of the original HDBSCAN-based system. Values are means over three seeds; higher clean accuracy and lower ASR/loss are better.

S1a. FLTrust extension on Fashion-MNIST under main poisoning attacks.

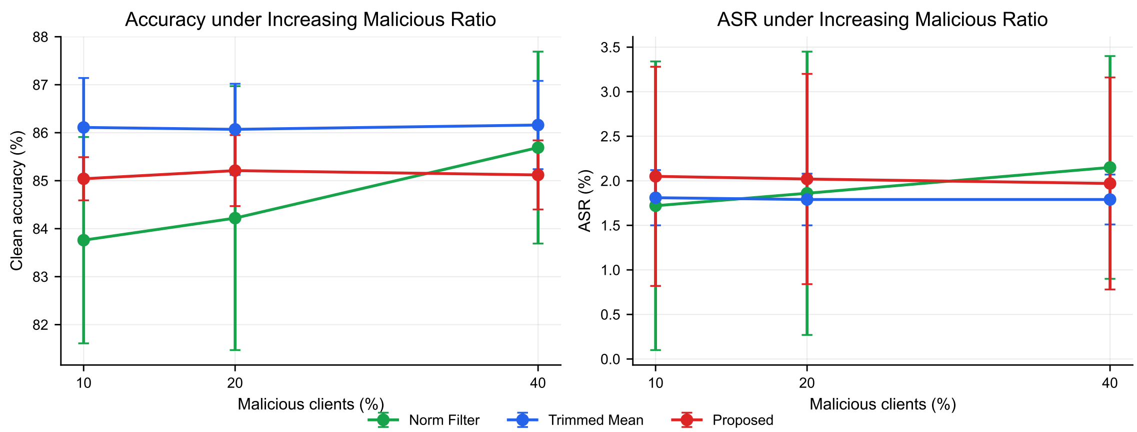
Dataset	IID	Attack	Method	Clean acc.	Acc. std	ASR	ASR std
Fashion-MNIST	No	Adaptive scaling	FLTrust	65.33	1.75	4.03	0.53
Fashion-MNIST	No	Backdoor	FLTrust	64.39	4.37	4.24	1.63
Fashion-MNIST	No	Sign flip	FLTrust	64.73	4.06	3.76	1.74
Fashion-MNIST	Yes	Adaptive scaling	FLTrust	73.62	0.57	3.06	0.39
Fashion-MNIST	Yes	Backdoor	FLTrust	72.77	0.55	4.21	0.71
Fashion-MNIST	Yes	Sign flip	FLTrust	73.56	0.46	3.05	0.36

S1b. Backdoor extension comparing FedAvg, FLTrust, FLAME-style defense, norm filtering, and the proposed method.

Dataset	IID	Attack	Method	Clean acc.	Acc. std	ASR	ASR std
CIFAR-10	No	Backdoor	FedAvg	71.41	5.68	7.97	5.72
CIFAR-10	No	Backdoor	FLAME-style	29.98	20.09	69.28	31.62
CIFAR-10	No	Backdoor	FLTrust	36.49	7.56	16.11	8.92
CIFAR-10	No	Backdoor	Norm Filter	78.92	1.19	2.71	0.89
CIFAR-10	No	Backdoor	Proposed	77.38	0.63	2.88	2.46
CIFAR-10	Yes	Backdoor	FedAvg	78.55	1.94	6.43	0.86
CIFAR-10	Yes	Backdoor	FLAME-style	70.54	2.16	7.85	2.49
CIFAR-10	Yes	Backdoor	FLTrust	23.98	24.21	68.15	55.17
CIFAR-10	Yes	Backdoor	Norm Filter	84.43	0.19	1.22	0.15
CIFAR-10	Yes	Backdoor	Proposed	83.50	0.89	1.22	0.45
Fashion-MNIST	No	Backdoor	FedAvg	84.93	1.64	90.27	8.52
Fashion-MNIST	No	Backdoor	FLAME-style	80.47	2.98	14.05	4.65
Fashion-MNIST	No	Backdoor	FLTrust	64.37	4.36	4.24	1.63
Fashion-MNIST	No	Backdoor	Norm Filter	84.59	2.22	2.78	0.37
Fashion-MNIST	No	Backdoor	Proposed	83.63	0.95	3.17	0.91
Fashion-MNIST	Yes	Backdoor	FedAvg	85.13	0.38	73.56	3.90
Fashion-MNIST	Yes	Backdoor	FLAME-style	78.44	5.00	7.20	2.48
Fashion-MNIST	Yes	Backdoor	FLTrust	72.74	0.56	4.21	0.81
Fashion-MNIST	Yes	Backdoor	Norm Filter	88.36	0.41	1.58	0.29
Fashion-MNIST	Yes	Backdoor	Proposed	88.33	0.30	1.52	0.31

Supplementary Figure S1. Malicious-ratio sensitivity

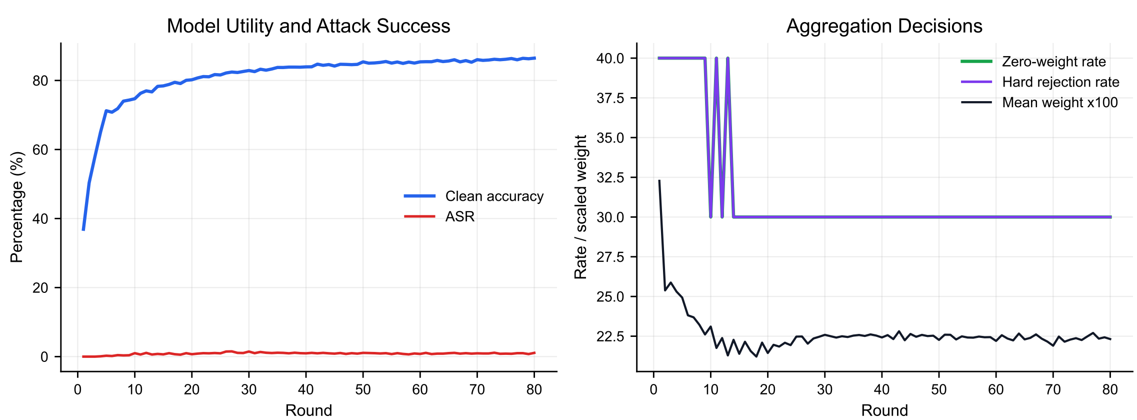
Figure S1 reports the auxiliary malicious-ratio sensitivity experiment on Fashion-MNIST Non-IID adaptive scaling. The figure is included as diagnostic evidence for robustness under increasing malicious participation; the main manuscript summarizes the result without treating it as a separate contribution.



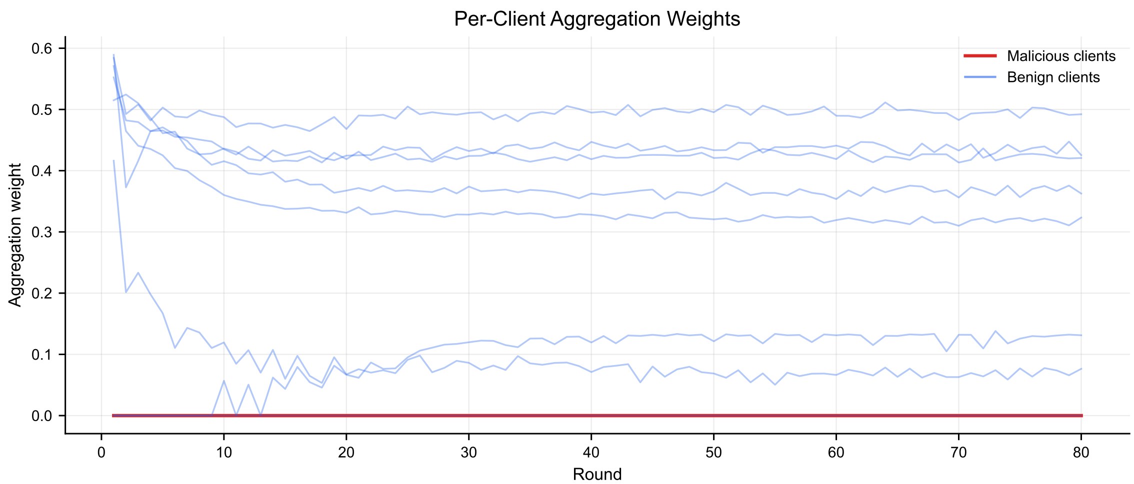
S1. Malicious-ratio sensitivity on Fashion-MNIST Non-IID adaptive scaling. Curves report three-seed means; higher accuracy and lower ASR are better.

Supplementary Figures S2–S3. Representative audit-case visualizations

Figures S2–S3 visualize the representative Fashion-MNIST Non-IID sign-flip audit case with the proposed method and seed 42. They complement the objective audit metrics in the main manuscript by showing the per-round trajectory and per-client weight assignment.



S2. Representative single-run audit trajectory. The figure shows clean accuracy, ASR, zero-weight rate, hard-rejection rate, and mean aggregation weight over 80 rounds.



S3. Per-client aggregation weights in the representative audit case. Malicious clients receive zero weight across all rounds, while benign clients retain nonzero contribution weights.

Supplementary Table S2. Coefficient sensitivity

Table S2 evaluates whether the proposed aggregation rule depends on one exact anomaly-score coefficient setting. The experiment uses Fashion-MNIST Non-IID under sign-flip and adaptive-scaling attacks. Values are means over three seeds.

S2. Coefficient sensitivity of the proposed method.

Attack	Variant	Clean acc.	Acc. std	ASR	ASR std
Adaptive scaling	Balanced	85.81	0.77	2.01	1.00
Adaptive scaling	Default	85.19	0.68	1.99	1.16
Adaptive scaling	Direction-heavy	86.30	1.00	1.95	0.94
Adaptive scaling	History-heavy	85.64	0.73	2.07	1.11
Adaptive scaling	Norm-heavy	83.04	2.94	2.02	1.07
Sign flip	Balanced	84.64	1.61	2.36	1.43
Sign flip	Default	84.10	1.96	2.25	1.47
Sign flip	Direction-heavy	84.08	2.20	2.31	1.47
Sign flip	History-heavy	84.78	1.47	2.46	1.34
Sign flip	Norm-heavy	84.01	1.49	2.05	1.46

Supplementary Table S3. Validator-dropout liveness boundary

Table S3 reports validator-dropout sensitivity for threshold finality. Values are means over 13 representative proposed-method runs. Valid blocks finalize when available validators meet the signing threshold and fail when availability falls below threshold.

S3. Validator-dropout liveness boundary.

Validators	Threshold	Offline	Available	Valid finalization	Valid failure	Verification ms
5	3	0	5	100.00%	0.00%	1.315
5	3	1	4	100.00%	0.00%	1.056
5	3	2	3	100.00%	0.00%	0.795
5	3	3	2	0.00%	100.00%	0.540
7	5	0	7	100.00%	0.00%	1.841
7	5	1	6	100.00%	0.00%	1.573
7	5	2	5	100.00%	0.00%	1.312
7	5	3	4	0.00%	100.00%	1.056

Supplementary Table S4. Event-driven finality simulation parameters

Table S4 reports the fixed parameters used to generate the event-driven validator-finality simulation in the main manuscript. The simulation is an application-layer finality model over measured proposal sizes and validator-verification times; it is not a full PBFT, HotStuff, Tendermint, or Fabric network implementation.

S4. Event-driven finality simulation parameters.

Parameter	Value	Role in simulation
Trials per setting	500	Number of deterministic pseudo-random trials for each client, committee, network, dropout, Byzantine, and proposal-type setting.
Random seed	20260615	Base seed used to make the event simulation reproducible across runs.
Client counts	10, 100, 200	Synthetic client-evidence sizes selected from the validator-scalability sweep.
Committees	$(5, 3)$ and $(11, 7)$	Validator count and signing threshold pairs used for event-driven finality outcomes.
Bandwidth	10 and 100 Mbps	Link bandwidth used to compute proposal and vote transmission time from measured serialized sizes.
RTT	10, 50, and 100 ms	Round-trip time; one-way propagation is modeled as RTT/2 with propagation jitter.
Dropout rate	0, 0.1, 0.2, and 0.4	Independent probability that each validator is unavailable in a trial.
Byzantine validators	0, $q-1$, and q	Validator-fault levels for honest, within-threshold, and threshold-compromised cases.
Proposal types	Valid and invalid	Valid proposals should be accepted; invalid proposals should be rejected unless Byzantine signers reach threshold.
Verification jitter	Uniform $[0.85, 1.15]$	Multiplier applied to per-validator verification time derived from measured committee verification latency.
Propagation jitter	Uniform $[0.8, 1.2]$	Multiplier applied to one-way propagation delay for proposal delivery and vote return.
Timeout	2000 ms	Trial outcome is timeout if neither an accept nor reject threshold is reached before this bound.
Vote size	max(256, S_f/q) bytes	Per-validator vote-return size, where S_f is the measured finalized-record size and q is the threshold.

For a valid proposal, each online non-Byzantine validator returns an accepting vote after proposal transmission, propagation delay, local verification, and vote transmission. For an invalid proposal, honest validators return rejecting votes and Byzantine validators return accepting votes. A proposal finalizes when accepting votes reach the threshold q before timeout. An invalid proposal is rejected when rejecting votes reach q before timeout. If neither side reaches q, the outcome is timeout. This rule is why a setting such as 11 validators with threshold 7 and six Byzantine validators can time out on an invalid proposal: six Byzantine accept votes are below threshold, while the five remaining honest reject votes are also below threshold.

Supplementary Table S5. Runtime and audit-log overhead

Table S5 reports the detailed runtime and audit-log overhead summarized in the main manuscript. Lower runtime is better; audit-log size reports the storage cost of keeping reconstructable decision evidence.

S5. Runtime and audit-log overhead summary for latest formal runs.

Dataset	Method	Runs	Round time (s)	Defense time (s)	Audit log (KB)
Fashion-MNIST	Norm Filter	18	1.434	0.0059	349.28
Fashion-MNIST	Proposed	18	1.788	0.0178	513.59
CIFAR-10	Norm Filter	9	18.550	0.0550	522.76
CIFAR-10	Proposed	9	18.719	0.0797	771.21

Supplementary Table S6. Estimated finality latency

Table S6 estimates finality latency under representative network settings using measured proposal sizes and validator-verification times. The estimate is an application-layer model, not a packet-level network simulator.

S6. Estimated finality latency under representative network settings.

Clients	Validators	Bandwidth Mbps	RTT ms	Proposal KB	Verification ms	Finality ms
10	5	100	10	12.55	2.05	23.15
100	5	100	10	112.36	18.04	47.33
200	5	100	10	223.66	35.13	73.53
100	5	10	50	112.36	18.04	210.89
200	5	10	50	223.66	35.13	319.15
100	11	100	10	112.36	39.25	68.59
100	11	10	50	112.36	39.25	232.66
200	11	10	100	223.66	77.82	462.41

Supplementary Table S7. Collusive-direction malicious-ratio sensitivity

Table S7 reports the auxiliary malicious-ratio sensitivity experiment under collusive-direction poisoning on Fashion-MNIST Non-IID. The result is used to identify the operating boundary of the defense under coordinated attackers. Proposed remains competitive at 10% malicious participation and has the best mean clean accuracy at 20%, but all evaluated methods collapse near random accuracy at 40%. This boundary result is reported in the supplementary material because it qualifies the robustness claim rather than forming the main contribution.

S7. Collusive-direction malicious-ratio sensitivity on Fashion-MNIST Non-IID. Values are mean ± standard deviation over three seeds; higher clean accuracy and lower ASR/loss are better.

Malicious ratio	Method	Runs	Clean acc.	Best acc.	ASR	Loss
0.1	FedAvg	3	84.38 ± 2.63	84.39	2.83 ± 0.46	0.4060
0.1	Norm Filter	3	69.67 ± 7.83	69.82	1.59 ± 1.39	0.8857
0.1	Proposed	3	82.44 ± 3.12	82.66	2.17 ± 1.04	0.4589
0.1	Trimmed Mean	3	83.07 ± 2.79	83.16	2.66 ± 0.76	0.4377
0.2	FedAvg	3	76.40 ± 7.84	76.40	2.79 ± 0.15	0.6711
0.2	Norm Filter	3	60.45 ± 4.70	63.42	1.95 ± 3.08	3.1285
0.2	Proposed	3	83.00 ± 2.50	84.63	1.79 ± 0.78	0.4527
0.2	Trimmed Mean	3	76.37 ± 0.39	76.59	2.37 ± 1.75	0.5569
0.4	FedAvg	3	24.87 ± 25.70	40.30	0.39 ± 0.68	358.5324
0.4	Norm Filter	3	10.01 ± 0.01	31.35	0.00 ± 0.00	834.9249
0.4	Proposed	3	10.00 ± 0.00	16.98	0.00 ± 0.00	2024.6780
0.4	Trimmed Mean	3	18.55 ± 14.80	19.11	0.10 ± 0.18	47.9982

Supplementary Table S8. Committee-capture probability under random validator sampling

Table S8 reports a simple committee-capture calculation for the threshold-finality assumption. Let ρ be the adversarial fraction in a permissioned validator candidate pool. If a committee of size m is sampled independently from that pool and a proposal finalizes with threshold q, the probability that the sampled committee contains at least q adversarial validators is

$$P_{\text{capture}} = \sum_{k=q}^{m} \binom{m}{k} \rho^k (1-\rho)^{m-k}.$$

This calculation is not a replacement for validator governance, Sybil resistance, or stake economics. It is included to make the security-performance trade-off explicit: larger committees and higher thresholds reduce threshold-capture probability under random sampling, but they increase proposal dissemination and vote-collection cost.

S8. Probability that adversarial validators reach the signing threshold under random committee sampling. Values are percentages.

Adversarial pool fraction ρ	$(m,q)=(5,3)$	$(m,q)=(7,5)$	$(m,q)=(11,7)$
0.10	0.856%	0.018%	0.002%
0.20	5.792%	0.467%	0.197%
0.30	16.308%	2.880%	2.162%
0.40	31.744%	9.626%	9.935%